Tech News

Technology Updates

Electronic glove offers human-like features for prosthetic hand users

The e-glove can be worn over a prosthetic hand to provide human-like softness, warmth, appearance and sensory perception

People with hand amputations experience difficult daily-life challenges, often leading to lifelong use of prosthetic hands and services.

An electronic glove, or e-glove, developed by Purdue University researchers can be worn over a prosthetic hand to provide human-like softness, warmth, appearance and sensory perception, such as the ability to sense pressure, temperature and hydration.

While a conventional prosthetic hand helps restore mobility, the new e-glove advances the technology by offering realistic human hand-like features for daily activities and life roles, with the potential to improve the wearers' mental health and well-being by helping them more naturally integrate into social contexts.

The e-glove uses thin, flexible electronic sensors and miniaturised silicon-based circuit chips on the commercially-available nitrile glove. It is connected to a specially-designed wristwatch, allowing for real-time display of sensory data and remote transmission to the user for post-data processing.

Chi Hwan Lee, assistant professor at Purdue's College of Engineering, in collaboration with other researchers at Purdue, University of Georgia and University of Texas, worked on the development of the e-glove technology. Lee says, "We developed a novel concept of the soft-packaged, sensor-instrumented e-glove built on a commercial nitrile glove, allowing it to seamlessly fit on arbitrary hand shapes. The e-glove is configured with a stretchable form of multimodal sensors to collect such information as pressure, temperature, humidity and electro-physiological bio-signals, while simultaneously providing realistic human hand-like softness, appearance and even warmth."

Lee and his team hope that the appearance and capabilities of the e-glove will improve the well-being of prosthetic hand users by allowing them to feel more comfortable in social contexts. The glove is available in different skin tone colours, has life-like fingerprints and artificial fingernails.

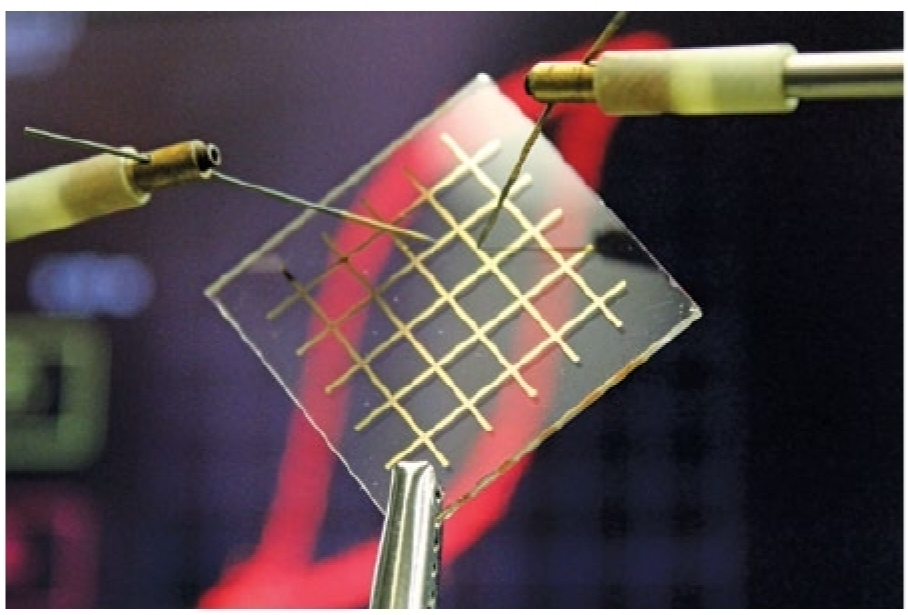
The fabrication process of the e-glove is cost-effective and manufacturable in high volume, making it an affordable option for users, unlike other emerging technologies with mind, voice and muscle control embedded within the prosthetic at a high cost. Additionally, these emerging technologies do not provide the human-like features that the e-glove provides.

Scientists develop nylon for transparent electronic devices

Scientists at Max Planck Institute for Polymer Research (MPI-P), led by Dr Kamal Asadi, have solved a four-decade-long challenge of producing very thin nylon films that can be used, for instance, in electronic memory components. These thin nylon films are several hundred times thinner than human hair and could, thus, be used for applications in bendable electronic devices or for electronics in clothing.

Nylons—a family of synthetic polymers—were first introduced in the 1940s for women's stockings and are now among the most widely-used synthetic fibres in textiles. These consist of a long chain of repeated molecular units, that is, polymers, where each repeat unit contains a specific arrangement of hydrogen, oxygen and nitrogen with carbon atoms.

Besides use in textiles, some nylons also exhibit ferroelectric properties, which means that positive and negative electric charges can be separated, and this state can



Transparent nylon could be an important building block for the development of transparent electronic circuits in the future (Credit: MPI-P)